AI WebShield

A privacy-first browser safety prototype for junior secondary students

FIRST-ROUND PROTOTYPE | CHROME MANIFEST V3 | LOCAL DEMO ONLY

THE PROBLEM

Students may share personal information in forms or trust a convincing sign-in page without noticing warning signs.

WebShield adds a calm pause before a form is submitted and gives a reason when a local demo page looks suspicious.

DESIGN PRINCIPLE

Help students check and choose. Do not secretly collect, change, or report what they type.

TARGET USERS

Junior secondary students learning safer browsing habits, with a teacher or mentor guiding the demo.

TWO PAUSE POINTS

- **1 Form review**
  Detect patterns in supported non-password text; suggest a reversible mask.
- **2 Page signal**
  Neutral, yellow caution, or red warning with a plain-language reason.

HOW THE PROTOTYPE WORKS

Local page -> local rule checks -> clear student choice

A **Student opens a fictional localhost sample**

B **Page metadata or supported text patterns are checked in the browser**

C **WebShield explains the signal and offers a next step**

D **Student reviews, masks, undoes, cancels, or continues**

EXPLAINABLE PAGE STATES

NEUTRAL	No demo signal found; not a safety guarantee.
CAUTION	Unfamiliar sample asks for sign-in/contact details.
HIGH	One-character lookalike plus sensitive-form metadata, or seeded fixture.

IMPLEMENTATION

Chrome MV3 extension; no backend, network service, threat feed, or broad site permission in this first demo.

PRIVACY, EVIDENCE & NEXT STEPS

- **Privacy by design**
  Password values are skipped. Pattern checks run locally. Entered text is not transmitted or logged. Masking is proposed and reversible.
- **Proportional warnings**
  An unfamiliar form alone is yellow, not red. Strong red is limited to explainable demo evidence.
- **Evaluation status**
  Synthetic test scenarios are documented. Interactive Chrome QA is still pending; no accuracy results are claimed.
- **Known limits**
  The rules can miss threats or produce false alarms. The lookalike rule is a teaching fixture, not general phishing detection.
- **Next steps**
  Run the manual cases in Chrome, record passes/misses without personal data, refine wording, then test with mentor feedback.